

MAKAUT B.TECH

PROJECT MANAGEMENT AND ENTREPRENEURSHIP

HSMC 701 | Exam-Oriented Study Notes

How to use this note: Read the definitions first, learn the tables and answer frames, then practise the model questions. Examples are illustrative; reproduce the concepts in your own words in the examination.

Coverage

This document covers exactly the following prescribed components:

- Serial 1: Meaning and Concept of Entrepreneurship; Innovation and Entrepreneurship; Contributions of entrepreneurs to society; risk-opportunities perspective and mitigation of risks.
- Serial 2: Entrepreneurship - An Innovation; Challenges of Innovation; Steps of Innovation Management; Idea Management System; Divergent vs Convergent Thinking; Qualities of a prospective Entrepreneur.

Serial 1: Introduction to Entrepreneurship

Learning outcome: Explain entrepreneurship as a value-creating process; connect it with innovation; discuss its social contribution; and evaluate risk, opportunity and risk-mitigation choices.

1. Meaning and Concept of Entrepreneurship

Entrepreneurship is the process through which an individual or a team identifies an opportunity, mobilises resources, takes calculated risk and creates value by establishing or improving an economic or social venture. The value may be profit, employment, a technical solution, social benefit or a combination of these.

An entrepreneur is not merely a person who starts a business. The entrepreneur converts an idea into an organised activity, makes decisions under uncertainty and accepts responsibility for results. Entrepreneurship therefore combines opportunity recognition, innovation, resource organisation and risk management.

Term	Exam-ready meaning
Entrepreneur	A person who identifies an opportunity, organises resources and assumes calculated risk to create value.
Entrepreneurship	The process of creating, developing and managing a venture or solution by exploiting an opportunity.
Enterprise	The business or organised undertaking created and operated to deliver goods, services or solutions.
Intrapreneur	An employee who behaves entrepreneurially within an existing organisation.

Core elements of entrepreneurship

- Opportunity recognition: noticing an unmet need, inefficiency or emerging demand.
- Innovation: introducing a new or improved product, process, service, business model or market approach.
- Resource mobilisation: arranging people, funds, technology, materials, information and networks.
- Calculated risk-taking: acting despite uncertainty after assessing possible loss and return.
- Value creation: delivering benefits to customers, society, employees and the economy.
- Execution and growth: converting the idea into a viable, sustainable and scalable activity.

Engineering example: A team notices that small factories lose energy because motors run during idle periods. It develops a low-cost IoT sensor and dashboard that switches equipment to an efficient mode. The idea is technical; entrepreneurship appears when the team validates demand, builds a prototype, prices it, finds customers and manages the risk of launch.

Entrepreneurship as a process

1. Identify a problem or opportunity.
2. Generate and screen possible solutions.
3. Test feasibility: customer need, technical feasibility, cost and legal requirements.
4. Prepare a business model and resource plan.
5. Develop a prototype or minimum viable product (MVP).
6. Launch, obtain feedback, improve and scale.

2. Innovation and Entrepreneurship

Innovation means the practical application of a new idea that creates value. An invention becomes an innovation only when it is applied, adopted or used to solve a real problem. Entrepreneurship converts innovation into a viable offering and carries it to the market or to society.

Basis	Invention	Innovation	Entrepreneurship
Focus	New discovery or technical idea	Useful application of a new/improved idea	Building value around an opportunity
Main question	Can it be made?	Does it solve a problem better?	Can it be delivered sustainably?
Output	Prototype, formula or design	Adopted product/process/service	Venture, market solution or scalable activity
Example	A new battery chemistry	Battery pack with longer life and safe charging	Company supplying packs, service and recycling

Innovation may be product innovation (a new device), process innovation (a better method of manufacturing), service innovation (a new delivery experience), business-model innovation (a different way to earn revenue) or social innovation (a new solution to a social need).

3. Contributions of Entrepreneurs to Society

Entrepreneurs contribute beyond private profit. Their ventures influence economic development, technological progress and quality of life.

Contribution	How it helps	Illustration
Employment generation	Creates direct jobs and demand for suppliers, logistics and services.	A local EV-service start-up hires technicians and sources parts from nearby firms.
Innovation and productivity	Introduces improved products and processes that save time, cost or resources.	A predictive-maintenance tool reduces unexpected machine breakdowns.

Contribution	How it helps	Illustration
Economic growth	Adds output, income, investment and tax revenue.	A successful MSME expands production and sells to wider markets.
Regional development	Uses local skills/resources and reduces concentration in large cities.	A food-processing unit adds value to produce near farming areas.
Competition and choice	Encourages quality, fair price and better customer service.	New ed-tech firms improve access to skill training.
Social and environmental value	Addresses inclusion, health, waste, energy and community problems.	A venture converts e-waste into reusable components.

Exam point: Entrepreneurship can create both economic value (income and jobs) and social value (access, inclusion, sustainability and problem-solving).

4. Risk-Opportunities Perspective and Mitigation of Risks

An opportunity is a favourable situation in which a problem can be solved or a need can be served profitably or beneficially. Risk is the possibility that actual results will differ from expected results and may lead to loss. Entrepreneurs do not ignore risk; they assess, reduce, share, transfer or accept it after considering the expected opportunity.

Opportunity check	Question an entrepreneur should ask
Need	Is there a real and sufficiently important customer or social problem?
Customer	Who will use, pay for or support the solution?
Feasibility	Can the solution be built, delivered and maintained with available capability?
Value	Why is it better than existing alternatives?
Timing	Is the market, regulation and technology environment favourable now?
Return	Can the benefits justify the investment, effort and risk?

Risk type	Example	Practical mitigation
Market risk	Customers may not adopt the product.	Customer interviews, pilot launch, MVP and feedback.
Technical risk	Prototype may fail or not meet performance.	Testing, expert review, phased development and backup design.
Financial risk	Cash may run out before sales begin.	Budgeting, staged funding, break-even analysis and cash reserve.

Risk type	Example	Practical mitigation
Operational risk	Supplier delay or quality problem.	Multiple suppliers, quality checks, process documentation.
Legal/regulatory risk	Product may violate a standard or licence requirement.	Compliance review, approvals, records and insurance.
People risk	Key member may leave or lack capability.	Training, role clarity, documentation and succession plan.

Risk mitigation cycle

7. Identify: list possible events that may affect cost, time, quality, safety or sales.
8. Assess: estimate likelihood and impact; prioritise high-likelihood/high-impact risks.
9. Plan response: avoid, reduce, transfer, share or accept the risk.
10. Implement controls: assign owner, deadline and resources for each action.
11. Monitor and learn: review signals, update the plan and use feedback.

Illustrative case: A student team wants to sell a smart water-level controller. The opportunity is recurring water wastage in apartments. Instead of producing 1,000 units immediately, the team installs 10 pilot units, tests reliability, collects feedback and uses an approved enclosure. This reduces market, technical and compliance risk while preserving the opportunity.

Serial 2: Entrepreneurship as Innovation

Learning outcome: Explain why entrepreneurship is an innovation-driven activity, describe innovation management, use idea-management thinking and identify qualities required in a prospective entrepreneur.

5. Entrepreneurship - An Innovation

Entrepreneurship is often described as an innovation because it creates a new value proposition or a new way of delivering value. The novelty need not be a world-first invention. It may be a meaningful improvement in design, process, access, cost, delivery, customer experience or business model.

For example, an engineering graduate may not invent solar power, but may innovate by designing a pay-per-use solar cold-storage service for small farmers. The innovation lies in matching technology, customer need, financing and delivery in a useful way.

Ways in which entrepreneurship innovates

- New product or service: a solution with different features or performance.
- New process: a faster, safer, cheaper or more reliable way of producing or delivering.
- New market approach: reaching an underserved user group or geography.
- New business model: changing how value is created, delivered and captured.
- New social solution: addressing a public problem through an economically sustainable model.

6. Challenges of Innovation

Innovation is uncertain because a new solution must work technically, attract users, fit available resources and survive competition. The following challenges commonly arise:

Challenge	Why it occurs	Constructive response
Uncertain customer acceptance	Users may prefer familiar solutions.	Observe users, validate the problem and run pilots.
Resource constraints	Time, money, equipment and talent are limited.	Prioritise essentials; build an MVP; use partnerships.
Technical uncertainty	New design may fail at scale or in field conditions.	Prototype, test, iterate and document learning.
Resistance to change	Employees, customers or institutions may fear disruption.	Communicate benefits, train users and introduce in phases.
Intellectual property and imitation	A useful idea may be copied or rights may be unclear.	Use IP awareness, confidentiality and speed of execution.

Challenge	Why it occurs	Constructive response
Regulatory and ethical concerns	Safety, data, environment or standards may apply.	Design for compliance and conduct early review.
Commercialisation gap	A good prototype may lack pricing, channel or service support.	Develop a business model alongside the technology.

7. Steps of Innovation Management

Innovation management is the systematic process of moving from opportunity to a tested, adopted and improved solution. It combines creativity with disciplined planning and learning.

Step	What happens	Output
1. Scan opportunity	Study users, technology, trends and pain points.	Problem statement / opportunity area
2. Generate ideas	Create many possible solutions without early judgement.	Idea pool
3. Select and validate	Compare value, feasibility, cost, risk and fit.	Shortlisted concept
4. Develop concept	Define features, users, business model and requirements.	Concept note / plan
5. Prototype and test	Build a model or MVP; test with users and experts.	Evidence and learning
6. Implement / launch	Prepare operations, quality, marketing and support.	Market or field deployment
7. Review and improve	Measure adoption, performance and feedback.	Improved/scaled solution

Important distinction: Idea generation is only the beginning. Innovation management includes selection, testing, implementation and continuous improvement.

8. Idea Management System (IMS)

An Idea Management System is a structured mechanism through which an organisation or venture collects, records, evaluates, develops and implements ideas from employees, customers, partners or other stakeholders. It prevents useful ideas from being lost and makes innovation a repeatable activity rather than a chance event.

Essential features of an IMS

- Clear problem themes or challenge statements.
- Easy submission of ideas through a portal, form, meeting or innovation cell.
- A standard idea template: problem, proposed solution, expected benefit, resources and risks.
- Transparent evaluation criteria such as impact, feasibility, novelty, cost and strategic fit.
- Feedback to contributors, including reasons for selection or rejection.
- Pilot funding, mentoring and ownership for selected ideas.
- A record of outcomes and learning for future improvement.

Stage in IMS	Typical activity	Example
Capture	Students suggest ideas for reducing campus electricity waste.	Online form with problem, idea and estimated benefit.
Screen	Panel scores ideas against cost, feasibility and impact.	Sensor-based lighting idea ranks high.
Develop	Selected team receives mentor and small prototype budget.	Team builds occupancy-sensor prototype.
Pilot	Solution is tested in one corridor.	Energy use is compared before and after.
Scale / learn	Successful idea expands; failed idea yields learning.	Pilot is extended to other blocks.

9. Divergent and Convergent Thinking

Creative problem-solving requires two complementary modes of thinking. Divergent thinking opens up many possible choices; convergent thinking narrows those choices to the most suitable one.

Basis	Divergent thinking	Convergent thinking
Purpose	Generate many alternatives.	Select the best alternative.
Question	What are all possible ways?	Which option best meets the criteria?
Approach	Open, exploratory, imaginative and non-judgemental initially.	Analytical, evaluative, evidence-based and decision-oriented.
Useful tools	Brainstorming, mind maps, SCAMPER, observation.	Feasibility analysis, scoring matrix, testing, cost-benefit analysis.
Output	Broad idea pool.	Chosen concept or solution.
Example	List ways to cut hostel food waste.	Choose composting after comparing cost, space and participation.

Remember: Do not converge too early. First create enough options; then use criteria to select and

refine the strongest one.

10. Qualities of a Prospective Entrepreneur

A prospective entrepreneur need not be born with every quality. Most qualities can be developed through observation, training, small projects, feedback and disciplined practice.

Quality	Meaning in practice	How a B.Tech student can develop it
Opportunity orientation	Sees problems as possible value-creation opportunities.	Keep a problem diary and talk to users.
Initiative	Acts without waiting for perfect conditions.	Start small experiments or campus projects.
Creativity	Produces original or improved solutions.	Use brainstorming and cross-disciplinary learning.
Calculated risk-taking	Decides after evaluating downside and upside.	Prepare basic risk registers for projects.
Persistence	Continues despite setbacks and learns from failure.	Review failed prototypes and iterate.
Decision-making	Chooses timely action using evidence and priorities.	Use simple scoring matrices and deadlines.
Leadership and teamwork	Mobilises people and aligns effort.	Take responsibility in team projects.
Communication and networking	Explains value and builds useful relationships.	Present ideas; seek mentors and industry contacts.
Ethical responsibility	Considers safety, honesty, environment and stakeholders.	Check compliance, privacy and social impact early.
Learning orientation	Adapts based on data, users and changing conditions.	Seek feedback and update assumptions.

Rapid Revision: Serial 1 and 2

Keyword	One-line recall
Entrepreneurship	Opportunity + resources + calculated risk + value creation.
Innovation	Practical application of a new or improved idea that creates value.
Opportunity	A favourable, viable situation to solve a need or create value.
Risk mitigation	Identifying, assessing and reducing/handling possible loss.
Innovation management	Managing the journey from opportunity and ideas to adoption and

Keyword	One-line recall
	improvement.
Idea Management System	A formal process to capture, evaluate, develop and implement ideas.
Divergent thinking	Expanding the number of possible ideas.
Convergent thinking	Narrowing alternatives to the most suitable solution.

Likely Examination Questions with Model Answers

The answers below are concise models. In a 5- or 10-mark answer, add a brief introduction, headings, examples and a concluding line.

1. Define entrepreneurship. Explain its main elements.

Entrepreneurship is the process of identifying an opportunity, organising resources and taking calculated risk to create value through a venture or solution. Its main elements are opportunity recognition, innovation, resource mobilisation, calculated risk-taking, value creation and execution. For example, a team that identifies water wastage, develops a sensor-based controller, arranges funds and sells the solution is engaged in entrepreneurship.

2. Distinguish between invention, innovation and entrepreneurship.

An invention is a new idea, design or discovery. Innovation is its practical application to create value. Entrepreneurship is the organised process of carrying that value to users by arranging resources, managing risk and creating a sustainable venture. A new battery chemistry is an invention; a safe, useful battery pack is innovation; a firm that manufactures, sells and services it represents entrepreneurship.

3. How do entrepreneurs contribute to society?

Entrepreneurs generate employment, introduce innovations, raise productivity, contribute to income and tax revenue, develop regions, increase consumer choice and address social/environmental problems. Thus, their contribution includes both economic development and social value creation.

4. Explain the risk-opportunity perspective of entrepreneurship.

An entrepreneur evaluates an opportunity by checking customer need, feasibility, timing, value and expected return. At the same time, the entrepreneur identifies risks such as market, technical, financial, operational and legal risks. The correct approach is calculated risk-taking: reduce uncertainty through pilots, research, testing, budgeting, compliance and contingency plans rather than avoiding all action.

5. What is innovation management? State its steps.

Innovation management is the systematic management of ideas from opportunity identification to implementation and improvement. Its steps are: scanning opportunity; generating ideas; selecting and validating; developing the concept; prototyping and testing; implementing or launching; and reviewing and improving. It ensures that creativity is converted into a practical, adopted solution.

6. What is an Idea Management System?

An Idea Management System is a structured process for collecting, recording, evaluating, developing and implementing ideas from stakeholders. It uses clear themes, simple submission, evaluation criteria, feedback,

pilot support and learning records. It helps an organisation turn scattered suggestions into usable innovations.

7. Compare divergent and convergent thinking.

Divergent thinking generates many different possible solutions and is open, exploratory and non-judgemental at the start. Convergent thinking evaluates alternatives against criteria and selects the best one. In innovation, a team first diverges during brainstorming, then converges using feasibility, cost, customer value and risk analysis.

8. Discuss qualities of a prospective entrepreneur.

A prospective entrepreneur should have opportunity orientation, initiative, creativity, calculated risk-taking ability, persistence, decision-making ability, leadership, communication, networking, ethical responsibility and a learning orientation. These qualities enable the person to identify needs, mobilise resources, overcome setbacks and build trust while creating value.

Answer-writing Checklist

- Begin with a precise definition in 2-3 lines.
- Use headings or numbered points matching the question.
- Add one relevant engineering/business illustration.
- For compare questions, use a table with at least four bases.
- For process questions, state steps in correct order.
- End with a one-line conclusion linking the concept to value creation or practical implementation.

Final revision cue: Entrepreneurship is not blind risk-taking. It is innovation-driven value creation through opportunity recognition, resource organisation, calculated risk and continuous learning.